

A Quick Comparison of the UMBC, NOAA, and UW CrIS Calibration Algorithms

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Introduction

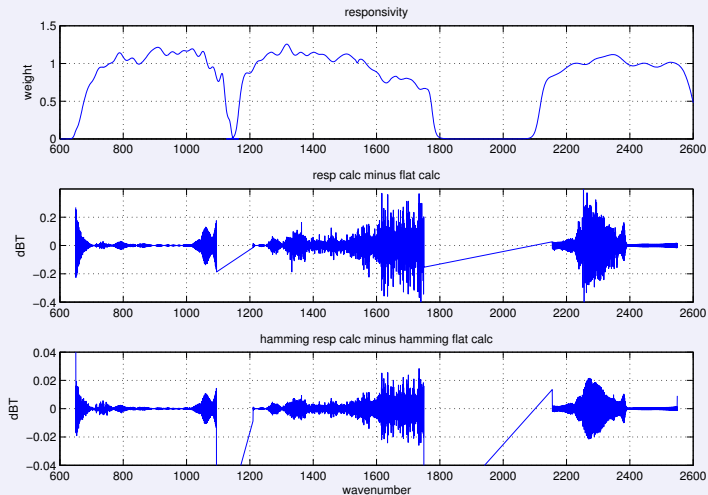
- ▶ We make several three-way comparisons of the UMBC, NOAA, and UW/NASA CrIS (Cross-track Infrared Sounder) calibrated radiance products, primarily for the J1/NOAA-20 sounder.
- ▶ These results are only spot checks of a few granules and FOVs, intended to look for obvious differences or problems. They were chosen as being representative of a larger number of tests.
- ▶ Overall, the three products look similar, at least for apodized radiances. This is reassuring. Different algorithms, and the presence or absence of a polarization correction, show up as measurable differences.
- ▶ Before showing differences, we take a quick look at reference truth and the polarization correction, the main causes of the differences.

- ▶ UMBC radiances are from UMBC CCAST versions V20 or V21, c. 2023, using the “c7” algorithm, except as noted.
- ▶ UW/NASA radiances are from their L1b V2 product, except as noted.
- ▶ NOAA radiances are from the NOAA CrIS SDR production system, as of late 2023 and early 2024.
- ▶ the CCAST c7 algorithm uses a “ratio-first” calibration algorithm, periodic sinc at the decimated sensor grid for the ILS, cosine apodization at the band edges of the sensor grid, resampling to take the sensor to the user grid, and a sinc basis for reference truth.
- ▶ NOAA and UW use an “SA-first” algorithm, a sinc basis for the ILS, with an extended SA matrix, circular shift of the extended user grid, resampling, and reference truth convolved with instrument responsivity. NOAA and UW V3 include a polarization correction, while UMBC and UW V2 do not.
- ▶ Fundamental calibration equations are given in the appendix.

Reference truth

- ▶ Reference truth is calculated upwelling radiance—what we expect the sounder to see for a particular atmospheric state.
- ▶ We generate this starting with kcarta, which gives an approximation to line-by-line calculations at an 0.0025cm^{-1} grid, and convolve the result to match instrument OPD, passbands, and filter specs.
- ▶ This is a conventional method for calculating reference truth. However NOAA and UW/NASA do something different—they convolve reference truth as above with instrument responsivity, and use this modified reference truth for instrument validation.
- ▶ There are problems with this. Mainly, it amounts to trying to define away an error, rather than fixing it. In addition, responsivity varies among the CrIS family of sounders, so that you have no single, even nominal, reference truth.
- ▶ As the following plot shows, the difference is significant for unapodized radiances, more than 0.2 K at many points.

Responsivity vs flat reference truth



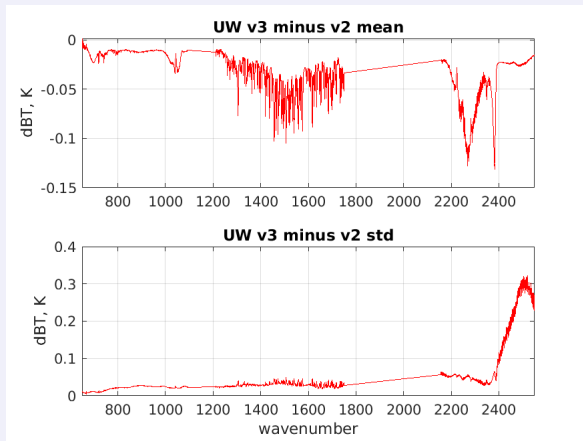
Validation

- ▶ The UMBC calibration algorithm was validated with simple “flat” reference truth, with a sinc basis, while NOAA and UW/NASA were validated with reference truth convolved with instrument responsivity.
- ▶ What this means in practice is that even if everything else was perfect, their respective calibrated radiance products would differ by the difference of responsivity vs flat reference truth, as shown above.
- ▶ This is a significant part of what we see in the unapodized differences that we show below.
- ▶ Note that most of the differences disappear with moderate apodization, as is often applied in practice. But there are cases where we would like to use the full resolution of the instrument.
- ▶ A list of UMBC validation presentations is given in the appendix. These (and in particular ccast_noaa3 and 4) show UMBC and NOAA are in good agreement with their respective values for reference truth. The reports are bundled with the ccast git repo.

Polarization correction

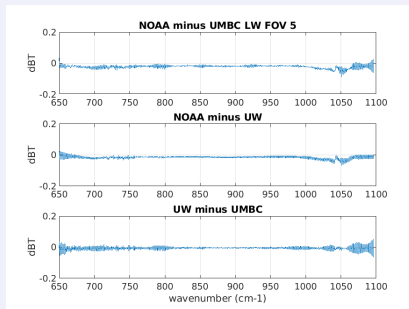
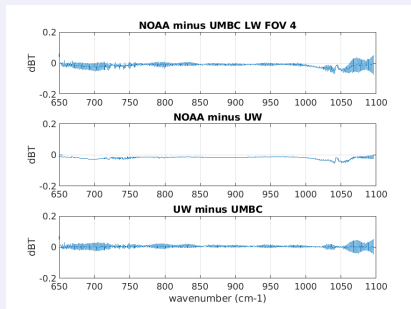
- ▶ We see the effect of different reference truth mostly at higher spectral resolutions (that is, at a finer frequency grid). These differences largely disappear with apodization.
- ▶ In contrast, the polarization correction gives rise to significant differences at lower spectral resolutions (a coarser grid) which persist with apodization.
- ▶ NOAA and UW V3 include a polarization correction, while UMBC and UW V2 do not.
- ▶ In the following slide, we show the polarization correction as the difference of UW v2 and v3 over all obs for 5 Oct 2023, for Hamming apodized radiances. Stats are done with brightness temps. std is the std of the differences, not the difference of separate v2 and v3 stds. In addition to the polarization correction, we see v3 has a significant cold bias, relative to v2.

Polarization correction



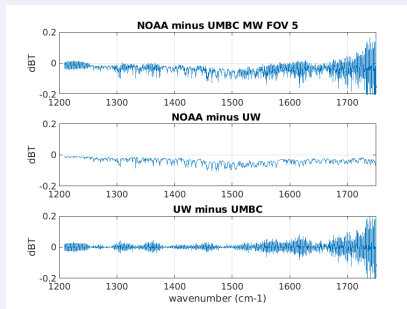
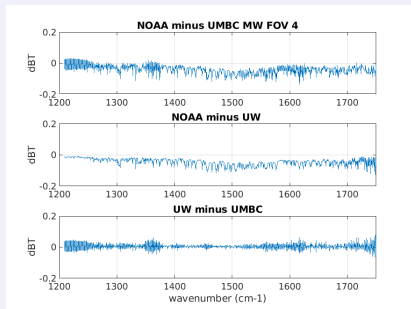
UW v3 minus v2 for 5 Oct 2023, for Hamming apodized radiances. Stats are done with brightness temp. std is the std of the differences, not the difference of separate v2 and v3 stds. In addition to the polarization correction, we see v3 has a significant cold bias, relative to v2.

LW unapodized comparisons



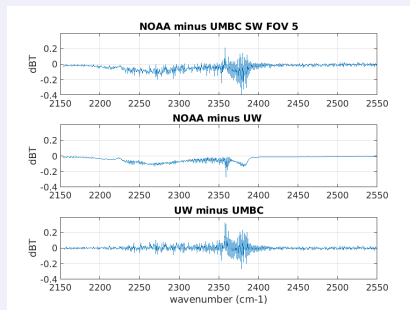
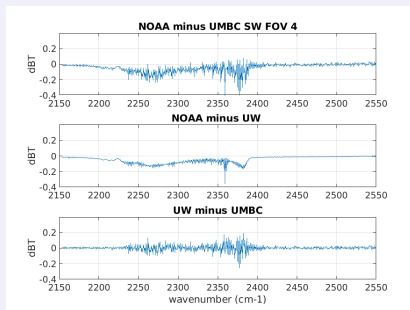
A 3-way comparison of NOAA, UMBC, and UW, LW unapodized radiances, for FOVs 4 and 5. Results are for J1/NOAA-20, 5 Oct 2023, UTC 19:03, lat 55 lon -97, 1200 matching obs, all FORs.

MW unapodized comparisons



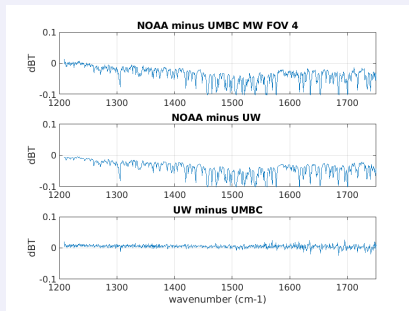
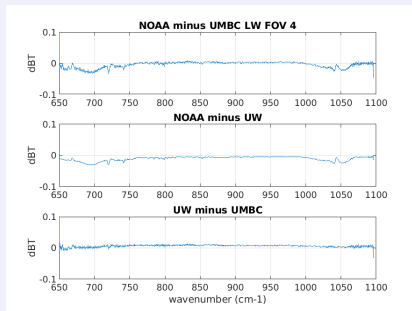
A 3-way comparison of NOAA, UMBC, and UW, MW unapodized radiances, for FOVs 4 and 5. Results are for J1/NOAA-20, 5 Oct 2023, UTC 19:03, lat 55 lon -97, 1200 matching obs, all FORs.

SW unapodized comparisons



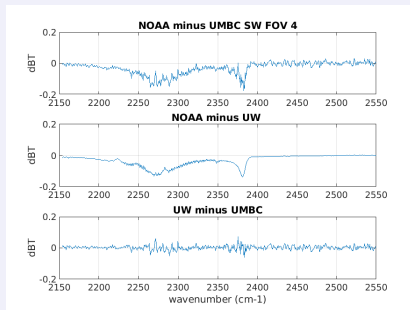
A 3-way comparison of NOAA, UMBC, and UW, SW unapodized radiances, for FOVs 4 and 5. Results are for J1/NOAA-20, 5 Oct 2023, UTC 19:03, lat 55 lon -97, 1200 matching obs, all FORs. Note the range of the Y axis has been extended to ± 0.4 .

LW & MW apodized comparisons



A 3-way comparison of NOAA, UMBC, and UW, LW and MW Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis has been reduced to ± 0.1 .

SW apodized comparisons

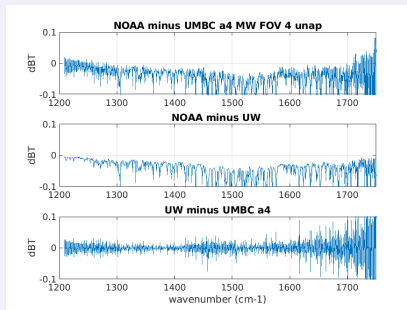
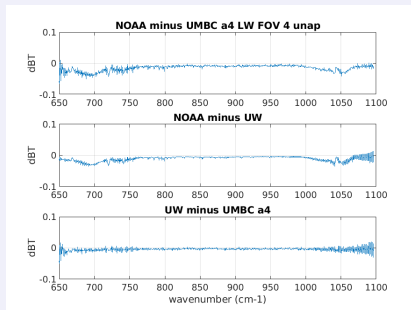


A 3-way comparison of NOAA, UMBC, and UW, SW Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.2K$ for this plot.

Discussion I

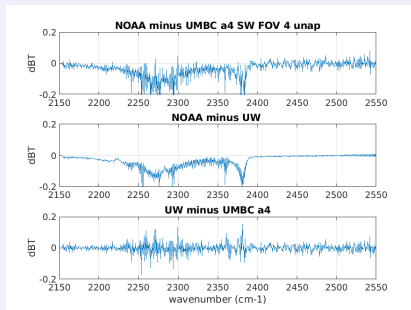
- ▶ The apodized differences between UW and UMBC are very small, for all three bands. Given the significant differences in the processing algorithms, this is reassuring.
- ▶ The unapodized differences between UW and UMBC appear to be mainly due to the different calibration algorithms and reference truth.
- ▶ The apodized differences between NOAA and UMBC, and NOAA and UW, appear to be due mainly to the polarization correction.
- ▶ The unapodized differences between NOAA and UMBC appear to be a mix of differences due to the calibration algorithms and polarization correction.
- ▶ The unapodized differences between NOAA and UW appear to be due to the polarization correction.
- ▶ In the following slides we compare NOAA and UW with the UMBC “a4” algorithm.

Unapodized a4 comparisons



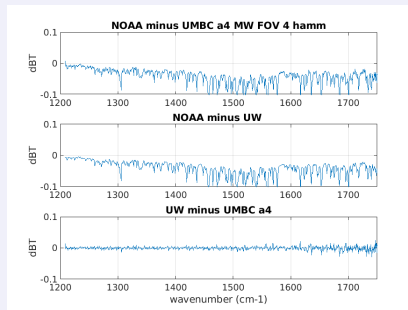
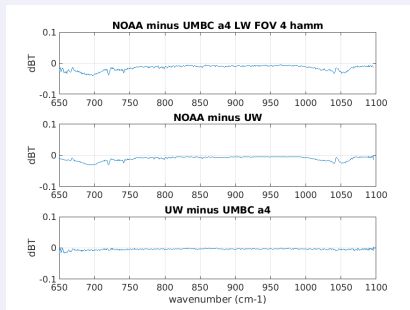
3-way comparison of NOAA, UMBC A4, and UW, LW and MW unapodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.1K$ for these plots.

Unapodized a4 comparisons



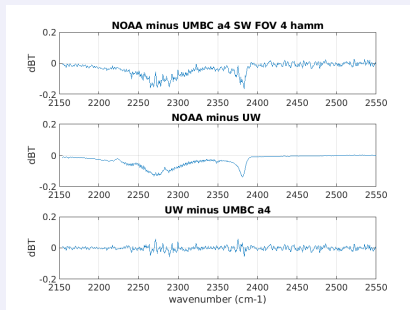
3-way comparison of NOAA, UMBC A4, and UW SW Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.2K$ for these plots.

Apodized a4 comparisons



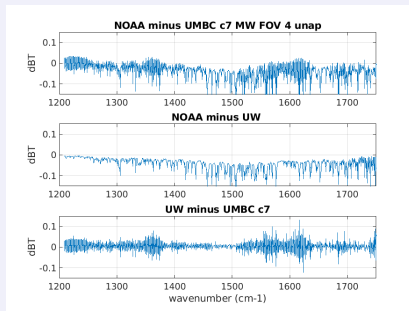
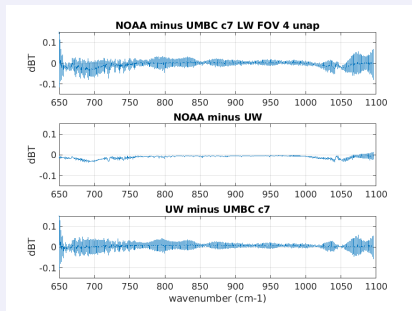
3-way comparison of NOAA, UMBC A4, and UW, LW and MW, Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.1K$ for these plots.

Apodized a4 comparisons



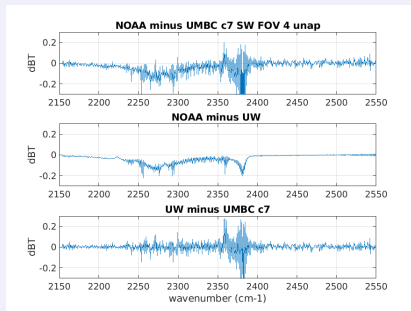
3-way comparison of NOAA, UMBC A4, and UW, SW Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.2K$ for these plots.

Unapodized c7 comparisons



3-way comparison of NOAA, UMBC c7, and UW, LW and MW unapodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note this is the same test method as the 5 Oct 2023 UTC 19:03 tests, but with different data. The range of the Y axis is $\pm 0.15K$ for these plots.

Unapodized c7 comparisons



3-way comparison of NOAA, UMBC c7, and UW, SW unapodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note this is the same test method as the 5 Oct 2023 UTC 19:03 tests, but with different data. The range of the Y axis is $\pm 0.3K$ for this plot.

Discussion II

- ▶ UMBC A4 is the UMBC implementation of the “SA first” algorithm used by NOAA and UW. In general these residuals are smaller, as we would expect. In this group of plots we are only showing FOV 4, for which the unapodized residuals were relatively large.
- ▶ The NOAA minus UMBC and UW minus UMBC unapodized residuals are similar in shape, but about half the size of the residuals from slides 9–11, where UMBC used the C7 algorithm. Note that these are for the same day but different granules.
- ▶ The NOAA minus UW unapodized residuals are almost identical to the residuals from slides 9–11. This is as we would expect, since the only change is that we are looking at different granules.
- ▶ The NOAA minus UMBC and UW minus UMBC apodized residuals are almost identical to the residuals from slides 12 and 13. Again this is to be expected, since the change is from UMBC C7 to A4, which has very little effect on apodized radiances.
- ▶ Finally, the unapodized c7 comparisons (slides 19–20) are very similar to the earlier unapodized c7 comparisons (slides 9–11), since the only difference is the data.

Conclusions

- ▶ We did a 3-way comparison of CrIS calibrated radiance products, in the form of spot checks for a few granules and FOVs, primarily for the J1/NOAA-20 sounder. Overall, the products look similar, at least for apodized radiances.
- ▶ Differences in unapodized radiances, up to a little more than 0.2 K, are mainly due to different calibration algorithms, validated with different versions of reference truth, and also in part due to the presence or absence of a polarization correction.
- ▶ Differences in apodized radiances, up to a little more than 0.1 K, are mainly due to the presence or absence of a polarization correction.
- ▶ Other differences are due to different calibration algorithms and other processing details. The validation reports, and in particular `ccast_noaa3`, `ccast_noaa4`, and `cris_psinc.pdf`, in the `ccast` repo, have more detail.

Future work

- ▶ Unfortunately, UMBC CCAST lacks a polarization correction, and a few other desirable features such as a Doppler correction. Development mostly stopped after around 2020, aside from apps for TVAC and post-launch validation, and there are no resources or clear demand for further work.
- ▶ Even lacking the polarization correction, CCAST may still be superior when an unapodized product or fast processing is desired. See the README in the CCAST git repo for a summary of the ways in which CCAST and the NOAA and UW products differ.
- ▶ We got much “mileage” out of CCAST over the years, processing and reprocessing the CrIS observations up to around 2023. We still use many of the libs and procedures for things like the J1-J4 TVAC analysis, focal plane fitting, validation after launch, and so on.
- ▶ We should have written a validation paper for CCAST. However the time for that was probably ten years ago.
- ▶ This presentation was reconstructed from Slack discussions from late 2023 and early 2024.

CCAST calibration equations

CCAST reference equation

$$r_{ES}^{user} = F \cdot r_{ICT}^{sensor} \cdot f_{cos} \cdot SA^{-1} \cdot f_{cos} \cdot \frac{\Delta ES}{\Delta IT} \quad (1)$$

$$r_{ES}^{user} = F \cdot f_{cos} \cdot SA^{-1} \cdot f_{cos} \cdot r_{ICT}^{fov} \cdot \frac{\Delta ES}{\Delta IT} \quad (2)$$

NOAA algorithm 4

$$r_{ES}^{user} = r_{ICT}^{user} \cdot \frac{F \cdot f_{ATBD} \cdot SA^{-1} \cdot f_{ATBD} \cdot \left(\frac{\Delta ES}{\Delta IT}\right) \cdot |\Delta IT|}{F \cdot f_{ATBD} \cdot SA^{-1} \cdot f_{ATBD} \cdot |\Delta IT|} \quad (3)$$

$$\Delta ES = ES - \langle SP \rangle, \quad \Delta IT = \langle IT \rangle - \langle SP \rangle$$

Calibration equation parameters

$$ES = \text{NLC}(\text{FIR}^{-1}(ES_0))$$

$$\langle IT \rangle = \text{NLC}(\text{FIR}^{-1}(\langle IT_0 \rangle))$$

$$\langle SP \rangle = \text{NLC}(\text{FIR}^{-1}(\langle SP_0 \rangle))$$

NLC is the ATBD non-linearity correction and ES_0 , IT_0 , and SP_0 are uncorrected earth-scene, ICT, and space spectra. $\langle IT_0 \rangle$ and $\langle SP_0 \rangle$ are moving averages, by default over 9 scans. FIR^{-1} is pointwise division by the spectral form of the numeric filter

- ▶ $r_{\text{ICT}}^{\text{user}}$ is expected ICT radiance at the user grid
- ▶ $r_{\text{ICT}}^{\text{sensor}}$ is expected ICT radiance at the sensor grid
- ▶ $r_{\text{ICT}}^{\text{fov}} = \text{SA} \cdot r_{\text{ICT}}^{\text{sensor}}$, expected uncorrected ICT radiance